A no-go theorem for non-multiplexed universality of state-conditioned IQP circuits

Supplementary theoretical material

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Abstract

We ask whether a single instantaneous quantum polynomial-time (IQP) model, whose circuit parameters are shared across conditions and whose condition appears only in a classically tractable input state, can be a universal conditional generator without using the condition as a selector for a bank of unconditional generators. It cannot. The obstruction is independent of the IQP gate set. Any fixed circuit followed by measurement is a quantum-to-classical channel, so the total-variation distance between two output laws cannot exceed the trace distance between their input states. Consequently, exact universality over conditional laws requires the input states for every two distinct conditions to have orthogonal supports. Uniform approximation of all continuous conditional laws by a resource-scaled family requires the corresponding states to become asymptotically perfectly distinguishable. Perfectly distinguishable condition states are operational labels and permit measurement followed by branch selection; this is precisely the routing resource excluded by a non-multiplexed notion of universality.

We also prove that a condition register which controls an IQP diagonal and is then discarded produces a classical mixture of condition-independent IQP branches. Thus selector and Bernstein constructions do not evade the obstruction; they implement multiplexing in coherent notation. For the paper's broadcast product encoder, we derive an explicit lower bound on the uniform approximation error. A target that changes from one deterministic visible string to another over an interval of width h has error at least $\frac{1}{2}(1-\sqrt{N}L_\alpha h/2)_+$ for N encoding qubits and an L_α-Lipschitz angle. At the reported Gaussian setting $N = 19$, $L_\alpha = 0.70\pi$, and $h = 0.01$, this lower bound is approximately 0.476. Fixed-width Fourier and parameter-dimension obstructions give separate counterexamples for the tested complete two-local models. Classical tractability of the inputs changes estimator cost, not these expressivity bounds.

1 Question and answer

The model in the accompanying paper uses one circuit for all conditions. A condition c changes an input state ρ_c; a fixed IQP unitary, hidden-qubit discarding, and computational-basis measurement produce a visible law $q_\theta(\cdot \mid c)$. The question addressed here is deliberately stronger than whether several unconditional IQP generators can be placed in orthogonal blocks and selected by c.

Definition 1.1 (Label separation and non-multiplexed universality)**.** Let $\mathcal{C}$ be a condition space and let $\rho_c^{(R)}$ denote the condition state at resource size R. The encoder family is *asymptotically label-separating* if

$$\sup_R D_{\mathrm{tr}}\left(\rho_c^{(R)}, \rho_{c'}^{(R)}\right) = 1 \quad \text{for every } c \neq c'. \tag{1}$$

We call it *non-multiplexed* if it is not label-separating, so at least one pair retains a trace-distance gap from one uniformly over resources. A model family is *universal* for a target class

$\mathcal{P}$ if, for every $p \in \mathcal{P}$ and every $\varepsilon > 0$, some resource size and some shared circuit parameters satisfy

$$\sup_{c \in \mathcal{C}} \mathrm{TVD}(q_\theta(\cdot \mid c), p(\cdot \mid c)) < \varepsilon. \tag{2}$$

The label-separation criterion formalizes the requested exclusion of "stitching together" condition-addressed IQP circuits. Sections 3 and 4 prove that universality for all conditional laws, or even for all total-variation-continuous laws on $[0, 1]$, contradicts this clause. The result is a counterexample to the broad universality claim:

> *Classically tractable conditional inputs do not make a shared IQP architecture conditionally universal. Exact universality forces orthogonal condition labels; asymptotic universality for all continuous kernels forces asymptotically orthogonal labels. Excluding that routing limit excludes the claimed universality.*

The conclusion does not rule out approximation of a narrower target class compatible with the encoding. It identifies a necessary compatibility inequality and quantifies its violation.

2 Model and distances

Let V contain $n \geq 1$ visible qubits, let H contain any number of hidden qubits, and let E contain any ancillary or condition-encoding qubits. The input state at condition $c \in \mathcal{C}$ is a density operator ρ_c on VHE. A fixed parameter choice θ specifies all subsequent operations: an IQP unitary, discarding of HE, and measurement of V. These operations define a quantum-to-classical channel

$$\mathcal{M}_\theta(\rho)(x) = \mathrm{Tr}(M_x^{(\theta)} \rho), \qquad x \in \{0, 1\}^n, \tag{3}$$

where $\{M_x^{(\theta)}\}_x$ is the input-picture positive-operator-valued measure (POVM). Hence

$$q_\theta(x \mid c) = \mathcal{M}_\theta(\rho_c)(x). \tag{4}$$

Nothing in the first no-go theorem depends on the IQP form of the channel. It therefore holds for arbitrary diagonals, arbitrary hidden width, and arbitrary trainable phase locality.

We take the encoding map to be fixed at each resource size, as in the paper. If the encoder has shared trainable parameters φ, one may replace the resource index R everywhere below by the pair (R, φ). All proofs are unchanged; the necessary suprema then range over every admissible encoder.

For probability laws p, q on a finite set and density operators ρ, σ, define

$$\mathrm{TVD}(p, q) = \frac{1}{2} \sum_x |p(x) - q(x)|, \tag{5}$$

$$D_{\mathrm{tr}}(\rho, \sigma) = \frac{1}{2} \|\rho - \sigma\|_1. \tag{6}$$

We use $(a)_+ = \max\{a, 0\}$. These distance facts are standard; a self-contained reference is Watrous [1]. Total variation also has the event characterization

$$\mathrm{TVD}(p, q) = \max_A |p(A) - q(A)|. \tag{7}$$

3 The contraction obstruction

Lemma 3.1 (Measurement contracts distinguishability)**.** *For every quantum-to-classical channel $\mathcal{M}$ and density operators ρ, σ,*

$$\mathrm{TVD}\big(\mathcal{M}(\rho), \mathcal{M}(\sigma)\big) \leq D_{\mathrm{tr}}(\rho, \sigma). \tag{8}$$

Proof. Write $\mathcal{M}(\rho)(x) = \mathrm{Tr}(M_x\rho)$. For an output event A, let $M_A = \sum_{x\in A} M_x$. The POVM conditions give $0 \le M_A \le I$. The variational characterization of trace distance is

$$D_{\mathrm{tr}}(\rho,\sigma) = \max_{0\le F\le I} |\,\mathrm{Tr}[F(\rho-\sigma)]|. \tag{9}$$

Using Eq. (7),

$$\begin{aligned}\mathrm{TVD}\big(\mathcal{M}(\rho),\mathcal{M}(\sigma)\big) &= \max_A |\,\mathrm{Tr}[M_A(\rho-\sigma)]| &(10)\\ &\le \max_{0\le F\le I} |\,\mathrm{Tr}[F(\rho-\sigma)]|, &(11)\end{aligned}$$

which is Eq. (8). □

The following inequality is the central quantitative result.

Theorem 3.2 (Pairwise approximation lower bound)**.** *For any two conditions c, c', any target conditional law p, and any fixed channel $\mathcal{M}_\theta$,*

$$\begin{aligned}&\max\big\{\mathrm{TVD}\big(q_\theta(\cdot\mid c), p(\cdot\mid c)\big), \mathrm{TVD}\big(q_\theta(\cdot\mid c'), p(\cdot\mid c')\big)\big\}\\ &\qquad\ge \frac{1}{2}\big[\mathrm{TVD}\big(p(\cdot\mid c), p(\cdot\mid c')\big) - D_{\mathrm{tr}}(\rho_c,\rho_{c'})\big]_+ .\end{aligned} \tag{12}$$

Proof. Set $e_c = \mathrm{TVD}(q_\theta(\cdot\mid c), p(\cdot\mid c))$, and similarly define $e_{c'}$. Two applications of the triangle inequality and Lemma 3.1 give

$$\begin{aligned}\mathrm{TVD}\big(p(\cdot\mid c), p(\cdot\mid c')\big) &\le e_c + \mathrm{TVD}\big(q_\theta(\cdot\mid c), q_\theta(\cdot\mid c')\big) + e_{c'} &(13)\\ &\le e_c + D_{\mathrm{tr}}(\rho_c,\rho_{c'}) + e_{c'} &(14)\\ &\le 2\max\{e_c, e_{c'}\} + D_{\mathrm{tr}}(\rho_c,\rho_{c'}). &(15)\end{aligned}$$

Rearranging, and using nonnegativity of the left-hand side of Eq. (12), proves the claim. □

Equation (12) is an expressivity certificate for any proposed conditional encoding. A target can separate two conditions no more sharply than their encoded states, up to twice the approximation error.

Corollary 3.3 (Exact universality requires orthogonal labels)**.** *Suppose a fixed input encoding and a trainable family of channels represent every conditional law on a condition set containing two distinct points. Then*

$$D_{\mathrm{tr}}(\rho_c,\rho_{c'}) = 1 \qquad \textit{for every } c \ne c'. \tag{16}$$

Equivalently, ρ_c and $\rho_{c'}$ have orthogonal supports.

Proof. For a selected pair $c \ne c'$, choose a target with $p(\cdot\mid c) = \delta_{0^n}$ and $p(\cdot\mid c') = \delta_{1^n}$. These laws have total-variation distance one. Exact representation makes the left-hand side of Eq. (12) zero, so $D_{\mathrm{tr}}(\rho_c,\rho_{c'}) \ge 1$. Trace distance between states is at most one, hence equality holds. Equality is equivalent to orthogonality of their supports. Repeating the argument for every pair gives the result. □

Pairwise orthogonal states can be distinguished with zero error by one measurement. A model supplied with such states therefore receives a classical condition label in operational form: it can measure the label and route computation to a condition-specific branch. Exact universality does not produce a genuinely shared conditional mechanism; it requires the multiplexing resource that the question excludes.

Corollary 3.4 (A classically tractable counterexample)**.** *If $\rho_c = \rho_0$ for every c, then $q_\theta(\cdot \mid c) = q_\theta(\cdot \mid c')$ for every pair of conditions and every parameter choice. The family cannot represent any target that differs between two conditions.*

The constant state may be a computational-basis product state. It is classically tractable under the strongest standard notions: it can be sampled exactly and its amplitudes can be evaluated exactly. Thus "classically tractable input states" alone is not a sufficient premise for conditional universality.

4 Resource scaling does not remove the obstruction

An asymptotic universality statement may choose a new input width and circuit for each requested accuracy. Index these resources by R and write the condition state as $\rho_c^{(R)}$.

Theorem 4.1 (Necessary asymptotic separation)**.** *Let $\mathcal{C} = [0, 1]$, and suppose a resource-scaled family is universal in the sense of Eq. (2) for every conditional law $p(\cdot \mid c)$ that is continuous in total variation. Then, for every $c \neq c'$,*

$$\sup_R D_{\mathrm{tr}}\left(\rho_c^{(R)}, \rho_{c'}^{(R)}\right) = 1. \tag{17}$$

More precisely, approximation with uniform error ε of a suitable continuous target requires

$$D_{\mathrm{tr}}\left(\rho_c^{(R)}, \rho_{c'}^{(R)}\right) \geq 1 - 2\varepsilon. \tag{18}$$

Proof. Assume without loss of generality that $c < c'$. Define a continuous function $\lambda : [0, 1] \to [0, 1]$ that equals zero on $[0, c]$, is linear from zero to one on $[c, c']$, and equals one on $[c', 1]$. The target

$$p(\cdot \mid t) = (1 - \lambda(t))\delta_{0^n} + \lambda(t)\delta_{1^n} \tag{19}$$

is continuous in total variation and its endpoint laws have distance one. Apply Theorem 3.2 to a resource R that approximates this target with uniform error ε. Rearrangement yields Eq. (18). Sending ε to zero gives Eq. (17). □

The theorem permits a different resource R for different targets and accuracies. It still forces every fixed pair of distinct conditions to become arbitrarily distinguishable somewhere in the family.

Definition 4.2 (Uniform overlap preservation)**.** A resource-scaled encoder is uniformly overlap preserving if either of the following sufficient conditions holds:

1. there are $c \neq c'$ and $\gamma > 0$ such that $D_{\mathrm{tr}}(\rho_c^{(R)}, \rho_{c'}^{(R)}) \leq 1 - \gamma$ for every R; or
2. there is a common modulus ω with $\omega(\delta) \to 0$ as $\delta \to 0$ and

$$D_{\mathrm{tr}}(\rho_c^{(R)}, \rho_{c'}^{(R)}) \leq \omega(|c - c'|) \quad \text{for every } R, c, c'. \tag{20}$$

Corollary 4.3 (No-go for overlap-preserving families)**.** *Under condition 1, the ramp target in Eq. (19) has uniform approximation error at least $\gamma/2$. Under condition 2, there is a total-variation-continuous ramp target whose uniform approximation error is at least $1/4$ for every resource size and every circuit.*

Proof. The first statement follows immediately from Eq. (12). For the second, choose $h > 0$ with $\omega(h) \leq 1/2$, take two points at distance h, and use the continuous ramp between them. Theorem 3.2 gives $\frac{1}{2}(1 - \omega(h)) \geq 1/4$. □

Therefore, a universal resource-scaled family must lose every continuity bound of the form Eq. (20) that is uniform over resource size. Its condition encoding must sharpen with resources until nearby conditions act as nearly orthogonal labels. This is label amplification, not a shared smooth conditional representation.

5 Why discarded IQP selectors are classical multiplexers

One might try to evade the preceding conclusion by preparing a selector in superposition, applying a single diagonal unitary, and discarding the selector. The next proposition shows that this notation implements a classical mixture of branches.

Let S be a selector register and R collect all retained and hidden model registers. Consider the state immediately before the IQP diagonal. Assume that it factorizes as $\sigma_S(c) \otimes \tau_R$ and that only $\sigma_S(c)$ depends on c. Every unitary diagonal in the joint computational basis has the controlled form

$$D = \sum_{s\in\{0,1\}^r} |s\rangle\langle s|_S \otimes D_s, \tag{21}$$

where each D_s is diagonal on R.

Proposition 5.1 (Discarded-selector mixture identity)**.** *After applying D and discarding S, the state on R is*

$$\mathrm{Tr}_S\Big[D(\sigma_S(c) \otimes \tau_R)D^\dagger\Big] = \sum_{s\in\{0,1\}^r} w_c(s) D_s \tau_R D_s^\dagger, \qquad w_c(s) = \langle s|\sigma_S(c)|s\rangle. \tag{22}$$

Consequently, after any fixed channel on R, with S acted on only locally and then discarded, the visible measurement obeys

$$q(x \mid c) = \sum_s w_c(s) q_s(x), \tag{23}$$

where q_s is a condition-independent branch law. Off-diagonal coherence of $\sigma_S(c)$ has no effect on the output.

Proof. Expand $\sigma_S(c) = \sum_{s,t} \sigma_{st}(c)|s\rangle\langle t|$. Equation (21) gives

$$D(\sigma_S(c) \otimes \tau_R)D^\dagger = \sum_{s,t} \sigma_{st}(c)|s\rangle\langle t| \otimes D_s \tau_R D_t^\dagger. \tag{24}$$

Since $\mathrm{Tr}_S(|s\rangle\langle t|) = \delta_{st}$, only terms with $s = t$ survive, which proves Eq. (22). Linearity of the remaining channel proves Eq. (23). A final unitary acting only on S, including the final IQP Hadamards, cannot change its partial trace and therefore does not alter the identity. □

The identity covers finite one-hot selectors, binary-address registers, and Bernoulli selectors whose weights form Bernstein polynomials. Such constructions can be written as one large diagonal, but they remain a bank of q_s distributions joined by condition-dependent classical weights. Exact branch selection further requires distinguishable selector states, in agreement with Corollary 3.3. These constructions are valid compilations, but they are not a general non-multiplexed solution.

The factorization assumption is essential to this structural identity and is satisfied by the usual product selector constructions; the first IQP Hadamards preserve it. An encoder that entangles S and R need not obey Eq. (23), but it still obeys the contraction bound in Theorem 3.2.

6 Specialization to product rotation encoders

The paper uses tractable product inputs generated by Y rotations. Consider the general form

$$|\Phi(c)\rangle = \bigotimes_{j=1}^{N} R_Y(\alpha_j(c))|0\rangle, \qquad R_Y(a)|0\rangle = \cos(a/2)|0\rangle + \sin(a/2)|1\rangle. \tag{25}$$

The shared first Hadamard layer of an IQP circuit does not change pairwise overlaps. For pure states,

$$D_{\mathrm{tr}}(|\psi\rangle, |\phi\rangle) = \sqrt{1 - |\langle\psi|\phi\rangle|^2}. \tag{26}$$

Here and below, ket arguments of D_{tr} are shorthand for the corresponding rank-one projectors. This follows by diagonalizing the rank-two Hermitian operator $|\psi\rangle\langle\psi| - |\phi\rangle\langle\phi|$, whose two nonzero eigenvalues are $\pm\sqrt{1 - |\langle\psi|\phi\rangle|^2}$.

Using $\langle 0|R_Y(a)^\dagger R_Y(b)|0\rangle = \cos((a-b)/2)$ gives the exact distance

$$d_N(c, c') := D_{\mathrm{tr}}(|\Phi(c)\rangle, |\Phi(c')\rangle) = \sqrt{1 - \prod_{j=1}^{N} \cos^2\left(\frac{\alpha_j(c) - \alpha_j(c')}{2}\right)}. \tag{27}$$

For a broadcast encoder $\alpha_j(c) = \alpha(c)$,

$$d_N(c, c') = \sqrt{1 - \cos^{2N}\left(\frac{\alpha(c) - \alpha(c')}{2}\right)}. \tag{28}$$

At every fixed N, inputs with $|\alpha(c) - \alpha(c')| < \pi$ are nonorthogonal, so Corollary 3.3 rules out exact conditional universality.

Assume now that α is L_α-Lipschitz. Write $u = \cos^2(\Delta\alpha/2)$. The elementary inequalities $1 - u^N \le N(1-u)$ and $|\sin x| \le |x|$ imply

$$d_N(c, c') = \sqrt{1 - u^N} \le \sqrt{N(1-u)} \tag{29}$$

$$= \sqrt{N}\left|\sin\frac{\Delta\alpha}{2}\right| \le \frac{\sqrt{N}L_\alpha}{2}|c - c'|. \tag{30}$$

Corollary 6.1 (Sharp-transition counterexample)**.** *Let a continuous target change linearly from δ_{0^n} to δ_{1^n} across a condition interval of width h. Every fixed channel fed by the N-qubit broadcast encoder satisfies*

$$\sup_c \mathrm{TVD}\big(q_\theta(\cdot \mid c), p(\cdot \mid c)\big) \ge \frac{1}{2}\left(1 - \frac{\sqrt{N}L_\alpha h}{2}\right)_+. \tag{31}$$

Proof. The target laws at the two endpoints have total-variation distance one. Combine Eq. (30) with Theorem 3.2. □

For the reported Gaussian experiment, $N = 19$ and $\alpha(c) = \pi(0.35 - 0.70c)$, so $L_\alpha = 0.70\pi$. Taking $h = 0.01$ in Eq. (31) gives

$$\frac{1}{2}\left(1 - \frac{\sqrt{19}(0.70\pi)(0.01)}{2}\right) \approx 0.4760. \tag{32}$$

This bound holds even if the IQP diagonal is replaced by an arbitrary quantum circuit with unlimited trainable parameters.

For fixed $c \neq c'$ with $0 < |\Delta\alpha| < \pi$, Eq. (28) approaches one as $N \to \infty$. Repetition therefore evades a fixed pairwise bound only by making the product states asymptotically orthogonal. Simultaneously, Eq. (30) shows that the best common local modulus scales as $\sqrt{N}$. This is the label-amplification limit identified in Theorem 4.1.

7 Two independent fixed-architecture obstructions

The contraction theorem applies to arbitrary trainable channels. The IQP and finite-parameter structure impose additional restrictions on the fixed architectures tested in the paper.

7.1 Finite Fourier bandwidth

For the broadcast state in Eq. (25), the one-qubit density operator is

$$R_Y(\alpha)|0\rangle\langle 0|R_Y(\alpha)^\dagger = \frac{1}{2}\left(I + \sin(\alpha)X + \cos(\alpha)Z\right). \tag{33}$$

Every entry of the N-qubit product density operator is therefore a trigonometric polynomial of degree at most N. For any fixed POVM effect M_A,

$$q_\theta(A \mid \alpha) = \mathrm{Tr}(M_A|\Phi(\alpha)\rangle\langle\Phi(\alpha)|) \tag{34}$$

is therefore a real trigonometric polynomial of degree at most N. This is the standard Fourier-spectrum restriction induced by repeated angle encoding [2, 3].

Proposition 7.1 (Explicit Fourier counterexample)**.** *Fix N broadcast input rotations and let $0 < \eta \leq 1$. On a full 2π period, define a binary-supported target by*

$$p(0^n \mid \alpha) = \frac{1 + \eta\cos((N+1)\alpha)}{2}, \qquad p(1^n \mid \alpha) = 1 - p(0^n \mid \alpha). \tag{35}$$

Then every fixed downstream quantum channel obeys

$$\sup_\alpha \mathrm{TVD}\big(q_\theta(\cdot \mid \alpha), p(\cdot \mid \alpha)\big) \geq \frac{\eta}{4}. \tag{36}$$

Proof. Let $g(\alpha) = q_\theta(0^n \mid \alpha)$. It has zero Fourier coefficient at frequency $N+1$, whereas the coefficient of $f(\alpha) = p(0^n \mid \alpha)$ at that frequency is $\eta/4$. If $\|g - f\|_\infty < \eta/4$, then the magnitude of every Fourier coefficient of $g - f$ is smaller than $\eta/4$, contradicting its coefficient $-\eta/4$ at frequency $N+1$. Hence $\|g - f\|_\infty \geq \eta/4$. Total variation dominates the probability difference of the event $\{0^n\}$, which proves Eq. (36). □

The full-period assumption is needed only for the explicit constant $\eta/4$. On any closed interval I of nonzero length, the restrictions of degree-N trigonometric polynomials form a finite-dimensional, hence closed, proper subspace $\mathcal{T}_N$ of $C(I)$. The function in Eq. (35) is not in that subspace: equality on an interval would, by analyticity, imply equality on the full real line and contradict its nonzero $(N+1)$st Fourier coefficient. Consequently,

$$\delta_{N,I} := \inf_{g \in \mathcal{T}_N} \left\| g - \frac{1 + \eta\cos((N+1)\,\cdot)}{2} \right\|_{\infty,I} > 0. \tag{37}$$

The same binary-supported target therefore has approximation error at least $\delta_{N,I}$ on the paper's nondegenerate affine angle interval. Thus fixed bandwidth is not dense there either, although Eq. (36) gives the simpler numerical bound only on a full period.

7.2 Parameter dimension across finitely many conditions

Fix K encoded inputs and an n-bit visible output. Remove one redundant probability from each simplex, so the space of arbitrary conditional tables has dimension

$$d = K(2^n - 1). \tag{38}$$

An IQP ansatz with M periodic phase parameters defines a real-analytic map

$$F : \mathbb{T}^M \longrightarrow \mathbb{R}^d. \tag{39}$$

Proposition 7.2 (Dimension obstruction)**.** *If $M < d$, the image of F has Lebesgue measure zero in $\mathbb{R}^d$ and is not dense in the product of K probability simplices.*

Proof. The derivative of F has rank at most $M < d$ everywhere, so every point of $\mathbb{T}^M$ is critical. Sard's theorem implies that $F(\mathbb{T}^M)$ has d-dimensional Lebesgue measure zero. Because $\mathbb{T}^M$ is compact and F is continuous, the image is compact and therefore closed. A closed measure-zero subset cannot be dense in a simplex product with nonempty relative interior. □

For a complete two-local diagonal on Q qubits with one one-body and one two-body phase for each support,

$$M = Q + \binom{Q}{2}. \tag{40}$$

The reported settings give the following necessary-dimension failures. The Gaussian row uses its five-condition training grid; the Curie–Weiss row uses its six training temperatures.

Experiment	visible n	conditions K	phases M	$K(2^n - 1)$
Gaussian	19	5	190	2,621,435
Curie–Weiss	8	6	91	1,530

This count concerns exact coverage or density of arbitrary tables at the listed fixed inputs. It does not claim that the experimental targets require all table dimensions, nor does it replace the pairwise contraction bound.

8 Classical tractability is orthogonal to universality

The paper's classical training method [4] requires input states for which one can sample computational-basis strings and evaluate the amplitude ratios entering Pauli estimators. Product rotation states satisfy these requirements. So do constant product states, which give the immediate counterexample after Corollary 3.3, and orthogonal computational-basis labels, which enable routing. Tractability therefore includes examples at both extremes and does not control conditional expressivity.

More formally, tractability is an algorithmic property of a state description. The inequalities in Sections 3 and 4 depend only on the density operators themselves. No variance reduction, amplitude-ratio identity, or classical sampling oracle can increase $D_{\mathrm{tr}}(\rho_c, \rho_{c'})$, because the physical output is still obtained by the channel in Eq. (3).

Conversely, orthogonal selector states may be trivially classically tractable, while a generic condition-addressed diagonal can require an exponential number of phases to describe. Thus three resource questions must be kept separate:

1. whether the input admits the paper's classical estimator;
2. whether condition states remain overlapping or become routing labels;
3. whether the diagonal has a polynomial-size local description.

Recent unconditional IQP universality results concern the ability of a sufficiently general diagonal to reproduce one target law, sometimes with hidden qubits [5]. Compiling many such laws behind a selector is a useful construction, but Proposition 5.1 shows why it does not establish the shared conditional universality studied here.

9 Admissible positive statements and open boundary

The no-go result suggests the strongest possible target class for a fixed encoder. If a model approximates p with uniform error ε, then Theorem 3.2 necessarily gives, for all c, c',

$$\mathrm{TVD}\big(p(\cdot \mid c), p(\cdot \mid c')\big) \leq D_{\mathrm{tr}}(\rho_c, \rho_{c'}) + 2\varepsilon. \tag{41}$$

Therefore, a defensible positive theorem must restrict targets to kernels whose total-variation geometry is dominated by the trace-distance geometry of the chosen encoding. Even Eq. (41) is only necessary, not sufficient: locality, Fourier bandwidth, hidden width, and parameter sharing impose additional constraints.

For the complete two-local IQP family, it remains open whether a useful nontrivial subclass of encoding-compatible kernels admits a uniform approximation theorem with polynomial resources. Such a theorem would be a genuinely shared conditional result. It would have to exploit relations between nearby conditions rather than assign them to orthogonal blocks.

10 Conclusion

The requested broad, non-multiplexed universality statement is false. The proof is the channel-contraction inequality

$$\mathrm{TVD}(q_\theta(\cdot \mid c), q_\theta(\cdot \mid c')) \leq D_{\mathrm{tr}}(\rho_c, \rho_{c'}),$$

which yields an explicit approximation lower bound for every pair of conditions. Exact universality requires pairwise orthogonal condition states. Uniform universality for all continuous conditional laws requires asymptotic pairwise orthogonalization and the loss of every resource-uniform continuity modulus. These states are operational condition labels, so this route to broad universality implements the selector or routing behavior excluded in the question.

Discarded selectors do not provide a coherent loophole: an IQP diagonal turns them into the classical mixture in Eq. (23). The paper's fixed broadcast product encoder additionally admits the quantitative counterexample in Eq. (31), while finite Fourier support and insufficient parameter dimension independently rule out universality of the tested fixed-width complete two-local circuits. The appropriate claim is therefore a no-go theorem, together with the encoding-compatibility condition in Eq. (41), rather than a selector-based universality construction.

References

[1] John Watrous. *The Theory of Quantum Information*. Cambridge University Press, 2018. doi: 10.1017/9781316848142. URL https://cs.uwaterloo.ca/~watrous/TQI/.

[2] Maria Schuld, Ryan Sweke, and Johannes Jakob Meyer. The effect of data encoding on the expressive power of variational quantum-machine-learning models. *Physical Review A*, 103 (3):032430, 2021. doi: 10.1103/PhysRevA.103.032430.

[3] Berta Casas and Alba Cervera-Lierta. Multidimensional fourier series with quantum circuits. *Physical Review A*, 107(6):062612, 2023. doi: 10.1103/PhysRevA.107.062612.

[4] Erik Recio-Armengol, Shahnawaz Ahmed, and Joseph Bowles. Train on classical, deploy on quantum: Scaling generative quantum machine learning to a thousand qubits. arXiv:2503.02934, 2025. URL https://arxiv.org/abs/2503.02934.

[5] Andrii Kurkin, Kevin Shen, Susanne Pielawa, Hao Wang, and Vedran Dunjko. Universality and kernel-adaptive training for classically trained, quantum-deployed generative models. arXiv:2510.08476, 2025. URL https://arxiv.org/abs/2510.08476.